

The coppe document class

Version 3.5.1

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May 24, 2026

Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BIB_TE_X and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a mailing list where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain
`\mainmatter` three-parts: *front*, *main* and *back* matters [?]. Each one of these parts is started
`\backmatter` by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`.
The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Front cover

This element was recently introduced by the Academic Registry. It is automatically constructed by the `\maketitle` command.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation [?]. It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your
`\foreigntitle` monograph in the current and foreign languages. The default languages are Brazilian Portuguese and English. The `babel` package is automatically loaded by `coppe.cls`, so you do not need to load it again. The Brazilian Portuguese is the main language and the English is only required for the foreign abstract.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc.

`\examiner` students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

`\advisor{Title}{Advisor's Name}{Surname}{Degree}`

```
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract` (*env.*) As stated by the Academic Registry [?], abstracts must be in one page each, with `foreignabstract` (*env.*) at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.

`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which

`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is

`\printloabbreviations` provided, it will be used as sort key. This was later, in 2024, implemented also for

`\printlosymbols` `\abbrev`, otherwise `Symbol`, or `Abreviation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

COPPE_T_E_X produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no BIB_T_E_X `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` BIB_T_E_X styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib} % in the preamble
...
\printbibliography           % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard `LATEX` command used before all appendices. `COPPETEX` introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsceexam, msceexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msceexam`, and `dsceexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

english `CoppeTEX` uses `Babel`. The default language is Portuguese (actually `brazilian`), with English being the second language. If option `english` is used, English becomes the main language and Portuguese the secondary. Look at the `Babel` package to switch between languages.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

6 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and `pdflatex` will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package. It is also useful to enable hyperlinks. Fortunately, the `coppe` class will do this automatically if `hyperref` is loaded.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

`longquote` **Quotation** To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

7 A simple example

```
1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{booktabs}% tabelas mais bonitas
5 \usepackage{rotating}% rodando coisas, como tabelas
6 \usepackage{longtable}% tabelas longas
```

```

7 \usepackage[most]{tcolorbox} % caixas de texto
8 \usepackage{amsmath,amssymb}
9 \usepackage{hyperref}
10 \usepackage{listings} % para usar listagens
11
12 \addbibresource{example.bib}% base de referências (biblatex/biber)
13
14 \makelosymbols
15 \makeloabbreviations
16
17 \begin{document}
18   \title{Título da Tese}
19   \foreigntitle{Thesis Title}
20   \author{Nome do Autor}{Sobrenome}
21   \advisor{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
22   \advisor{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
23   \advisor{Prof.}{Nome do Terceiro Orientador}{Sobrenome}{D.Sc.}
24
25   \examiner{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
26   \examiner{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
27   \examiner{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
28   \examiner{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
29   \examiner{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
30   \department{PESC}
31   \date{01}{2024}
32
33   \keyword{Primeira palavra-chave}
34   \keyword{Segunda palavra-chave}
35   \keyword{Terceira palavra-chave}
36
37   \maketitle
38
39   \frontmatter
40   \dedication{A alguém cujo valor é digno desta dedicatória.}
41
42   \chapter*{Agradecimentos}
43
44   Gostaria de agradecer a todos.
45
46   \begin{abstract}
47
48   Apresenta-se, nesta tese, ...
49
50   \end{abstract}
51
52   \begin{foreignabstract}
53
54   In this work, we present ...
55
56   \end{foreignabstract}
57
58   \tableofcontents
59   \listoffigures
60   \listoftables

```

```

61 \printlosymbols
62 \printloabbreviations
63
64 \mainmatter
65 \chapter{Introdução}
66
67 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da
68
69 A classe \verb|coppe| foi criada por Vicente Helano e George Ainsworth, porém, em 2024, é
70
71 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.com}
72
73 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
74
75 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
76
77 É importante de notar que essa classe não foi construída sobre a classe \LaTeX \ para a A
78
79 Apesar desse modelo ser muito bom, ele tem um defeito: a limitação do sistema de referênci
80
81 Mais ainda, as regras da COPPE ainda não se adaptaram, no início de 2024, as novas regras
82
83 Este documento não substitui, mas complementa, o documento que descreve a classe.
84
85
86
87 \chapter{Configurações Iniciais}
88
89 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c
90
91 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
92
93 Recomendo ler o documento “The \verb*|coppe| document class” para entender melhor todas
94
95 \section{Linguagem principal do texto}
96
97 Essa classe considera que o texto principal está em português e algumas partes específicas
98 \begin{itemize}
99 \item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.
100 \item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
101 \end{itemize}
102
103 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe Coppe\T
104
105
106 \section{Por que usar o \LaTeX}
107
108 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso dess
109
110 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se p
111
112 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob
113
114 \section{Como e onde usar o \LaTeX}

```



```

115
116 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambientes
117 \begin{enumerate}
118 \item Na sua máquina, instalando uma versão completa como o MikTeX, típico do Windows, ou
119 \item Usar um ambiente na rede, como o Overleaf.
120 \end{enumerate}
121
122 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b
123
124
125 \chapter{Algumas Regras da COPPE}
126
127 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei
128
129 É imprescindível definir os símbolos, tal como o
130 conjunto dos números reais  $\mathbb{R}$  e o conjunto vazio  $\emptyset$ .
131 \syml{\mathbb{R}}{Conjunto dos números reais}
132 \syml{\emptyset}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como
133
134 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \ve
135
136 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice
137
138 \section{Citações}
139
140 Citações curtas podem ser feitas \quote{o comando quote} ou direto com “duas crases e dois
141
142 \begin{longquote}
143 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
144 feita com o ambiente \verb=longquote= The primary objective of this
145 investigation was to determine the feasibility of detecting corrosion in
146 aluminum Naval aircraft components with neutron radiographic interrogation
147 and the use of standard corrosion penetrameters. Secondary objectives
148 included the determination of the effect of object thickness on image quality,
149 the defining of minimum levels of detectability and a preliminary investigation
150 of a means whereby the degree of corrosion could be quantified with neutron
151 radiographic data. \cite{article-example}
152 \end{longquote}
153
154 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \verb*|na
155
156 Em todo caso, \textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em
157
158 \chapter{Floats}
159
160 Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \texti
161
162 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm
163
164 \textbf{Segundo a norma da ABNT, as legendas} \verb|\caption| \textbf{das figuras e quadros
165
166 Quadros são opcionais. Quando usados, tabelas passam a só conter números, enquanto quadros c
167
168

```

```

169 \section{Tabelas e Figuras Padrão}
170
171 Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.
172
173 \begin{table}[ht]
174 \centering % Centraliza a tabela
175 \caption{Exemplo de Tabela de Números}
176 \label{tab:exemplo_numeros}
177 \begin{tabular}{ccc} % Define a quantidade de colunas
178 \hline % Linha superior
179 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ % Cabeçalhos
180 \hline % Linha média
181 1 & 2 & 3 \\ % Primeira linha de dados
182 \hline
183 4 & 5 & 6 \\ % Segunda linha de dados
184 \hline
185 7 & 8 & 9 \\ % Terceira linha de dados
186 \hline
187 10 & 11 & 12 \\ % Quarta linha de dados
188 \hline % Linha inferior
189 \end{tabular}
190 \end{table}
191
192
193
194 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
195
196 \begin{figure}[ht]
197 \centering % Centraliza a figura
198 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
199 \caption{Exemplo de Figura com Legenda Abaixo} % Legenda da figura
200 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
201 \end{figure}
202
203
204
205
206
207 \section{Tabelas mais elegantes}
208
209 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Isso
210
211 \begin{table}[ht]
212 \centering % Centraliza a tabela
213 \caption{Exemplo de Tabela de Números mais elegantes}
214 \label{tab:exemplo_numerosbom}
215 \begin{tabular}{ccc} % Define a quantidade de colunas
216 \toprule % Linha superior
217 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ % Cabeçalhos
218 \midrule % Linha média
219 1 & 2 & 3 \\ % Primeira linha de dados
220 4 & 5 & 6 \\ % Segunda linha de dados
221 7 & 8 & 9 \\ % Terceira linha de dados
222 10 & 11 & 12 \\ % Quarta linha de dados

```

```

223 \bottomrule % Linha inferior
224 \end{tabular}
225 \end{table}
226
227 \section{Tabelas Longas ou Largas}
228
229 Se sua tabela é muito longa ou larga, existem várias opções.
230 \begin{itemize}
231     \item alterar o tamanho da letra
232     \item Usar o longtable
233     \item rodar a tabela, fazendo ela em \textit{landscape}
234     \item fazer a tabela dentro de um minibox
235 \end{itemize}
236
237
238 \subsection{Tabelas largas demais}
239
240 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
241
242 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
243
244 \begin{table}[ht]
245 \centering % Centraliza a tabela
246 \caption{Exemplo de Tabela Larga com Fonte Menor}
247 \label{tab:tabela_largafns}
248 \footnotesize % Aplica uma fonte menor para a tabela
249 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
250 \toprule
251 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} & \textbf{Coluna 4} & \textbf{Colu
252 \midrule
253 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
254 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
255 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
256 \bottomrule
257 \end{tabular}
258 \end{table}
259
260 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
261
262 \begin{table}[ht]
263 \centering
264 \caption{Exemplo de Tabela Redimensionada}
265 \label{tab:examplerb}
266 \resizebox{\textwidth}{!}{%
267 \begin{tabular}{llll}
268 \toprule
269 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
270 \midrule
271 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
272 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
273 \bottomrule
274 \end{tabular}%
275 }
276 \end{table}

```

```

277
278
279 Para rodar uma tabela muito larga em 90 graus no LaTeX, você pode usar o pacote \verb*|rotat
280
281 Aqui está um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
282
283 \begin{sidewaystable}
284 \centering
285 \caption{Sua Legenda Aqui}
286 \label{tab:sua_tabela}
287 \begin{tabular}{lll}
288 \toprule
289 Coluna 1 & Coluna 2 & Coluna 3 \\
290 \midrule
291 Item 1 & Item 2 & Item 3 \\
292 Item 4 & Item 5 & Item 6 \\
293 \bottomrule
294 \end{tabular}
295 \end{sidewaystable}
296
297 Se a tabela for muito longa, o ambiente \verb|longtable| é o ideal. Ele fornece comandos par
298
299 % Exemplo de tabela longa que se estende por várias páginas
300 \begin{longtable}{|c|c|c|}
301 % primeiro cabeçalho (é o caption)
302 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
303 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
304 \endfirsthead
305 % cabeçalho normal
306 \multicolumn{3}{c}%
307 {\table\ -- continuação da página anterior} \\
308 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
309 \endhead
310 % pé normal
311 \hline \multicolumn{3}{r|}{Continua na próxima página} \\ \hline
312 \endfoot
313 \hline
314 % último pé
315 \multicolumn{3}{r|}{Continua na próxima página} \\
316 \hline \hline
317 \endlastfoot
318
319 % Conteúdo da tabela
320 1 & 2 & 3 \\
321 4 & 5 & 6 \\
322 1 & 2 & 3 \\
323 4 & 5 & 6 \\
324 1 & 2 & 3 \\
325 4 & 5 & 6 \\
326 1 & 2 & 3 \\
327 4 & 5 & 6 \\
328 1 & 2 & 3 \\
329 4 & 5 & 6 \\
330 1 & 2 & 3

```

331 4 & 5 & 6 \\
 332 1 & 2 & 3 \\
 333 4 & 5 & 6 \\
 334 1 & 2 & 3 \\
 335 4 & 5 & 6 \\
 336 1 & 2 & 3 \\
 337 1 & 2 & 3 \\
 338 4 & 5 & 6 \\
 339 1 & 2 & 3 \\
 340 4 & 5 & 6 \\
 341 1 & 2 & 3 \\
 342 4 & 5 & 6 \\
 343 1 & 2 & 3 \\
 344 4 & 5 & 6 \\
 345 1 & 2 & 3 \\
 346 4 & 5 & 6 \\
 347 1 & 2 & 3 \\
 348 4 & 5 & 6 \\
 349 1 & 2 & 3 \\
 350 4 & 5 & 6 \\
 351 1 & 2 & 3 \\
 352 4 & 5 & 6 \\
 353 1 & 2 & 3 \\
 354 4 & 5 & 6 \\
 355 1 & 2 & 3 \\
 356 4 & 5 & 6 \\
 357 1 & 2 & 3 \\
 358 4 & 5 & 6 \\
 359 1 & 2 & 3 \\
 360 4 & 5 & 6 \\1 & 2 & 3 \\
 361 4 & 5 & 6 \\
 362 1 & 2 & 3 \\
 363 4 & 5 & 6 \\
 364 1 & 2 & 3 \\
 365 1 & 2 & 3 \\
 366 4 & 5 & 6 \\
 367 1 & 2 & 3 \\
 368 4 & 5 & 6 \\
 369 1 & 2 & 3 \\
 370 4 & 5 & 6 \\
 371 1 & 2 & 3 \\
 372 4 & 5 & 6 \\
 373 1 & 2 & 3 \\
 374 4 & 5 & 6 \\
 375 1 & 2 & 3 \\
 376 4 & 5 & 6 \\
 377 1 & 2 & 3 \\
 378 4 & 5 & 6 \\
 379 1 & 2 & 3 \\
 380 4 & 5 & 6 \\
 381 1 & 2 & 3 \\
 382 4 & 5 & 6 \\
 383 1 & 2 & 3 \\
 384 4 & 5 & 6 \\

```

385 1 & 2 & 3 \\
386 4 & 5 & 6 \\
387 1 & 2 & 3 \\
388 4 & 5 & 6 \\
389 1 & 2 & 3 \\
390 4 & 5 & 6 \\
391 1 & 2 & 3 \\
392 4 & 5 & 6 \\
393 1 & 2 & 3 \\
394 4 & 5 & 6 \\
395 1 & 2 & 3 \\
396 4 & 5 & 6 \\
397 1 & 2 & 3 \\
398 4 & 5 & 6 \\
399 1 & 2 & 3 \\
400 4 & 5 & 6 \\
401 1 & 2 & 3 \\
402 4 & 5 & 6 \\
403 1 & 2 & 3 \\
404 4 & 5 & 6 \\
405 1 & 2 & 3 \\
406 4 & 5 & 6 \\
407 1 & 2 & 3 \\
408 4 & 5 & 6 \\
409 1 & 2 & 3 \\
410 4 & 5 & 6 \\
411 1 & 2 & 3 \\
412 4 & 5 & 6 \\
413 1 & 2 & 3 \\
414 4 & 5 & 6 \\
415 1 & 2 & 3 \\
416 4 & 5 & 6 \\
417 1 & 2 & 3 \\
418 4 & 5 & 6 \\
419 1 & 2 & 3 \\
420 4 & 5 & 6 \\
421 1 & 2 & 3 \\
422 4 & 5 & 6 \\
423 1 & 2 & 3 \\
424 4 & 5 & 6 \\
425 1 & 2 & 3 \\
426 4 & 5 & 6 \\
427
428 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
429 \end{longtable}
430
431 \chapter{Revisão Bibliográfica}
432
433 Para ilustrar a completa adesão ao estilo de citação e listagem de
434 referências bibliográficas, a Tabela~\ref{tab:citation} apresenta citações
435 de alguns dos trabalhos contidos na norma fornecida pela CPGP da
436 COPPE, utilizando o estilo numérico. Tirando do comando inicial o parâmetro opcional numér
437
438 \begin{table}[h]

```

```

439 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
440 \texttt{\textbackslash cite} do \LaTeX\ e
441 o comando \texttt{\textbackslash citet},
442 fornecido pelo pacote \texttt{natbib}.}
443 \label{tab:citation}
444 \centering
445 {\footnotesize
446 \begin{tabular}{|c|c|c|}
447 \hline
448 Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
449 \hline
450 Livro & \cite{book-example} & \citet{book-example}\\
451 Artigo & \cite{article-example} & \citet{article-example}\\
452 Relatório & \cite{techreport-example} & \citet{techreport-example}\\
453 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
454 Anais de Congresso & \cite{inproceedings-example} &
455 \citet{inproceedings-example}\\
456 Séries & \cite{incollection-example} & \citet{incollection-example}\\
457 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
458 Dissertação de mestrado & \cite{mastersthesis-example} &
459 \citet{mastersthesis-example}\\
460 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
461 \hline
462 \end{tabular}}
463 \end{table}
464
465 \begin{table}[h]
466 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
467 \texttt{\textbackslash cite} do \LaTeX\ e
468 o comando \texttt{\textbackslash citet},
469 fornecido pelo pacote \texttt{natbib}. Além disso, usando o booktabs.}
470 \label{tab:citation1}
471 \centering
472 {\footnotesize
473 \begin{tabular}{ccc}
474 \toprule
475 Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
476 \midrule
477 Livro & \cite{book-example} & \citet{book-example}\\
478 Artigo & \cite{article-example} & \citet{article-example}\\
479 Relatório & \cite{techreport-example} & \citet{techreport-example}\\
480 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
481 Anais de Congresso & \cite{inproceedings-example} &
482 \citet{inproceedings-example}\\
483 Séries & \cite{incollection-example} & \citet{incollection-example}\\
484 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
485 Dissertação de mestrado & \cite{mastersthesis-example} &
486 \citet{mastersthesis-example}\\
487 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
488 \bottomrule
489 \end{tabular}}
490 \end{table}
491
492 \chapter{Alguns outros exemplo úteis}

```

```

493
494 \begin{tcolorbox}[title=Meu Textbox]
495 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
496 \end{tcolorbox}
497
498 \begin{tcolorbox}
499 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como
500 \end{tcolorbox}
501
502 \begin{figure}[ht]
503     \centering
504     \begin{tikzpicture}
505         \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{image.png}}
506         \begin{scope}[x={(image.south east)},y={(image.north west)}]
507             % Definindo o textbox dentro da figura
508             \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=
509                 \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
510                     Este textbox fala sobre como inserir um textbox dentro de uma figura usando
511                     \end{tcolorbox}
512                 ];
513             \end{scope}
514         \end{tikzpicture}
515         \caption{Figura com Textbox}
516         \label{fig:figura_com_textbox1}
517     \end{figure}
518
519
520 \begin{figure}[ht]
521     \centering
522     \begin{tcolorbox}
523 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como
524 \end{tcolorbox}
525     \caption{Figura com Textbox simples}
526     \label{fig:figura_com_textbox}
527 \end{figure}
528
529 \chapter{Método Proposto}
530 \chapter{Resultados e Discussão}
531
532 \section{Algumas Demonstrações}
533
534 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos  $\alpha$ 
535 \syml[beta]{Beta}{A palavra Beta mais e corrigida}
536 \syml[zzbeta]{\beta}{A letra  $\beta$  corrigida}
537 \syml{beta}{A palavra beta}
538 \syml{alpha}{A palavra alpha}
539 \syml[alpha]{Alpha}{A palavra Alpha}
540 \syml[zzalpha]{\alpha}{A letra  $\alpha$  corrigida}
541 \syml[marco]{Marco}{A palavra Marco corrigida}
542
543 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
544 \abbrev{GoT}{Game of Thrones}
545 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
546 \abbrev[iot]{IoT}{IoT ordenado como iot}

```



```

547 \abbrev[IOT]{IoT}{IoT ordenado como IoT}
548 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
549 \abbrev{IoT}{IoT com ordenação default}
550 \abbrev[ITU]{ITU}{ITU mesmo}
551
552
553
554
555 \chapter{Conclusões}
556
557 \backmatter
558
559 \printbibliography
560
561
562
563 \appendix
564
565 \chapter{Um apêndice}
566
567 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
568
569 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo
570
571
572 \chapter{Outro apêndice}
573
574 \annex
575
576
577 \chapter{Um Anexo}
578 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
579
580
581
582 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
583
584 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comand
585
586
587 \chapter{Outro Anexo}
588
589
590 \end{document}
591
592 \example

```

8 Implementation

8.1 The ‘coppe.cls’ file

```

593 \class
594 \def\filename{coppe.dtx}
595 \def\fileversion{v3.4}

```

```

596 \def\filedate{2024/01/24}
597 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
598 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
599 \LoadClass[12pt,a4paper,oneside]{book}
600 % The bibliography engine (biblatex+biber) is loaded further below, after babel,
601 % so the numbers class option can select the citation style at load time.
602 \RequirePackage{hyphenat}
603 \RequirePackage{lastpage}
604 \RequirePackage{ifthen}
605 \RequirePackage{graphicx}
606 \RequirePackage{setspace}
607 \RequirePackage{tabularx}
608 \RequirePackage{etoolbox}
609 \RequirePackage{eqparbox}
610 \RequirePackage{ltxcmds}
611 \RequirePackage[T1]{fontenc}
612 \RequirePackage[a4paper, bindingoffset=0.0cm, vcentering=true, %
613 top=2.5cm, bottom=2.5cm, left=3.0cm, right=3.0cm]{geometry}
614 \def\CoppeTeX{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
615 p\kern-.025em e}}\kern-.08em
616 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}

617 \newboolean{maledoc}
618 \setboolean{maledoc}{false}
619 %
620 % Class options.
621 % If you are writing a text in English, you must turn ‘English’ on.
622 % Otherwise, Portuguese is considered the main language.
623 \newif\if@english\@englishfalse
624 \DeclareOption{english}{\@englishtrue}
625 \DeclareOption{msc}{%
626   \newcommand{\@degree}{M.Sc.}
627   \newcommand{\@degree name}{Mestrado}
628   \newcommand{\local@degname}{Mestre}
629   \newcommand{\foreign@degname}{Master}
630   \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
631   \newcommand\foreign@doctype{Dissertation}
632 }
633 \DeclareOption{dsceexam}{%
634   \newcommand{\@degree}{D.Sc.}
635   \newcommand{\@degree name}{Doutorado}
636   \newcommand{\local@degname}{Doutor}
637   \newcommand{\foreign@degname}{Doctor}
638   \setboolean{maledoc}{true}
639   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
640   \newcommand\foreign@doctype{Qualifying Exam}
641 }
642 \DeclareOption{mscexam}{%
643   \newcommand{\@degree}{M.Sc.}
644   \newcommand{\@degree name}{Mestrado}
645   \newcommand{\local@degname}{Mestre}
646   \newcommand{\foreign@degname}{Master}
647   \setboolean{maledoc}{true}
648   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
649   \newcommand\foreign@doctype{Qualifying Exam}

```

```

650 }
651 \DeclareOption{dsc}{%
652   \newcommand{\@degree}{D.Sc.}
653   \newcommand{\@degreename}{Doutorado}
654   \newcommand{\local@degname}{Doutor}
655   \newcommand{\foreign@degname}{Doctor}
656   \newcommand\local@doctype{Tese}
657   \newcommand\foreign@doctype{Thesis}
658 }
659 \newif\if@coppenumeric\@coppenumericfalse
660 \DeclareOption{numbers}{\@coppenumerictrue}

```

Here is the default one-and-a-half line spacing. Users can change to double spacing by passing the `doublespacing` option.

```

661 \onehalfspacing
662 \DeclareOption{doublespacing}{%
663   \doublespacing
664 }

665 \ProcessOptions\relax
666 \if@english
667   \RequirePackage[brazilian,english]{babel}
668 \else
669   \RequirePackage[english,brazilian]{babel}
670 \fi
671 \addto{\captionsenglish}{%
672   \renewcommand{\bibname}{References}
673 }
674 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
675 % former natbib+bibtex setup. Loaded here, after babel, so its language strings
676 % map correctly. The numbers option selects the numeric style; otherwise the
677 % author-date default is used. natbib=true keeps \citel/\citep available.
678 \if@coppenumeric
679   \RequirePackage[backend=biber,datamodel=coppe,%
680     bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
681 \else
682   \RequirePackage[backend=biber,datamodel=coppe,%
683     bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
684 \fi
685 % Reference-list heading: a chapter titled with \bibname (so it still follows
686 % babel and the english option) and added to the table of contents, matching
687 % the previous thebibliography behaviour.
688 \defbibheading{bibliography}[\bibname]{%
689   \chapter*{#1}%
690   \addcontentsline{toc}{chapter}{#1}}
691 % The catalog page range (bib:begin/bib:end) was written by the old
692 % thebibliography redefinition, which biblatex's \printbibliography does not
693 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
694 % begin label uses that hook and the end label is emitted by wrapping
695 % \printbibliography to write it just after the reference list.
696 \AtBeginBibliography{\coppe@bibBegin}
697 \let\coppe@orig@printbibliography\printbibliography
698 \renewcommand\printbibliography[1][]{%
699   \coppe@orig@printbibliography[#1]%
700   \coppe@bibEnd}

```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

701 \newcommand\department[1]{%
702   \ifthenelse{\equal{#1}{PEB}}{
703     {\global\def\local@deptname{Engenharia Biom{\'e}dica}
704     \global\def\foreign@deptname{Biomedical Engineering}}{}
705   \ifthenelse{\equal{#1}{PEC}}{
706     {\global\def\local@deptname{Engenharia Civil}
707     \global\def\foreign@deptname{Civil Engineering}}{}
708   \ifthenelse{\equal{#1}{PEE}}{
709     {\global\def\local@deptname{Engenharia El{\'e}trica}
710     \global\def\foreign@deptname{Electrical Engineering}}{}
711   \ifthenelse{\equal{#1}{PEM}}{
712     {\global\def\local@deptname{Engenharia Mec{\'a}nica}
713     \global\def\foreign@deptname{Mechanical Engineering}}{}
714   \ifthenelse{\equal{#1}{PEMM}}{
715     {\global\def\local@deptname{Engenharia Metal{\'u}rgica e de Materiais}
716 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
717   \ifthenelse{\equal{#1}{PEN}}{
718     {\global\def\local@deptname{Engenharia Nuclear}
719     \global\def\foreign@deptname{Nuclear Engineering}}{}
720   \ifthenelse{\equal{#1}{PEN0}}{
721     {\global\def\local@deptname{Engenharia Oce{\'a}nica}
722     \global\def\foreign@deptname{Ocean Engineering}}{}
723   \ifthenelse{\equal{#1}{PPE}}{
724     {\global\def\local@deptname{Planejamento Energ{\'e}tico}
725     \global\def\foreign@deptname{Energy Planning}}{}
726   \ifthenelse{\equal{#1}{PEP}}{
727     {\global\def\local@deptname{Engenharia de Produ{\'c}o{\'e}o}
728     \global\def\foreign@deptname{Production Engineering}}{}
729   \ifthenelse{\equal{#1}{PEQ}}{
730     {\global\def\local@deptname{Engenharia Qu{\'i}mica}
731     \global\def\foreign@deptname{Chemical Engineering}}{}
732   \ifthenelse{\equal{#1}{PESC}}{
733     {\global\def\local@deptname{Engenharia de Sistemas e Computa{\'c}o{\'e}s}
734     \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
735   \ifthenelse{\equal{#1}{PET}}{
736     {\global\def\local@deptname{Engenharia de Transportes}
737     \global\def\foreign@deptname{Transportation Engineering}}{}
738   \ifthenelse{\equal{#1}{PENT}}{
739     {\global\def\local@deptname{Engenharia de Nanotecnologia}
740     \global\def\foreign@deptname{Nanotechnology Engineering}}{}
741 }

```

`\title` Used to enter the title in Brazilian Portuguese.

```

742 \renewcommand\title[1]{%
743   \global\def\local@title{#1}%
744 }

```

`\foreigntitle` Used to enter the foreign title.

```

745 \newcommand\foreigntitle[1]{%
746   \global\def\foreign@title{#1}%
747 }

```

`\advisor` Defines globally the title, name and academic degree of the advisors.

```
748 \newcount\@advisor\@advisor0
749 \newcommand\advisor[4]{%
750   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
751   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
752   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
753   \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
754   \global\advance\@advisor by 1
755   \ifnum\@advisor>1
756     \renewcommand\local@advisorstring{Orientadores}
757     \renewcommand\foreign@advisorstring{Advisors}
758   \fi
759 }
```

`\examiner`

```
760 \newcount\@examiner\@examiner0
761 \newcommand\examiner[3]{%
762   \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
763   \global\advance\@examiner by 1
764 }
```

`\author` It was redefined to allow the identification of the author's first names and surname.

```
765 \renewcommand\author[2]{%
766   \global\def\@authname{#1}
767   \global\def\@authsurn{#2}
768 }
```

`\date` This code makes easy to switch from dates in different languages.

```
769 \renewcommand\date[2]{%
770   \month=#1
771   \year=#2
772 }
```

`\local@monthname`

```
773 \newcommand\local@monthname{\ifcase\month\or
774   Janeiro\or Fevereiro\or Mar{\c c}o\or Abril\or Maio\or Junho\or
775   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}
```

`\foreign@monthname`

```
776 \newcommand\foreign@monthname{\ifcase\month\or
777   January\or February\or March\or April\or May\or June\or
778   July\or August\or September\or October\or November\or December\fi}
```

`\keyword`

```
779 \newcounter{keywords}
780 \newcommand\keyword[1]{%
781   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
782   \global\addtocounter{keywords}{1}
783 }
```

`\freeconfig` This command allows easy changing of core class parameters.

```
784 \newcommand\freeconfig[6]{%
```

```

785 \providecommand\@degree{}
786 \renewcommand\@degree{#1}
787 \providecommand\@degreename{}
788 \renewcommand\@degreename{#2}
789 \providecommand\local@degname{}
790 \renewcommand\local@degname{#3}
791 \providecommand\foreign@degname{}
792 \renewcommand\foreign@degname{#4}
793 \providecommand\local@doctype{}
794 \renewcommand\local@doctype{#5}
795 \providecommand\foreign@doctype{}
796 \renewcommand\foreign@doctype{#6}%
797 }
798 % \end{macrocode}
799 % \end{macro}
800 %
801 % \begin{macro}{\frontmatter}
802 % The number of pages for both frontmatter and mainmatter printed
803 % in the cataloging details page is computed by means of simple
804 % LaTeX labels.
805 % \begin{macrocode}
806 \renewcommand\frontmatter{%
807 \cleardoublepage
808 \@mainmatterfalse
809 \pagenumbering{roman}
810 \thispagestyle{empty}
811 \setcounter{page}{2}
812 \makefrontpage
813 \clearpage
814 \pagestyle{plain}
815 \ifthenelse{\boolean{maledoc}}{\}{\makecatalog}%
816 }

\mainmatter
817 \renewcommand\mainmatter{%
818 \coppe@mainBegin
819 \cleardoublepage
820 \@mainmattertrue
821 \pagestyle{plain}
822 \pagenumbering{arabic}}

\backmatter
823 \renewcommand\backmatter{%
824 \if@openright
825 \cleardoublepage
826 \else
827 \clearpage
828 \fi}
829 %

\maketitle
830 \renewcommand\maketitle{%
831 \pagenumbering{alph}
832 \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%

```

```

833 \begin{titlepage}
834 \begin{flushleft}
835 \vspace*{1.5mm}
836 \setlength\baselineskip{0pt}
837 \setlength\parskip{1mm}
838 \makebox[20mm][c]{\hspace{4.8cm}\includegraphics{coppe-logo}}
839 \end{flushleft}
840 \vspace{1.05cm}
841 \begin{center}
842 \nohyphens{%
843   \if@english
844     \MakeUppercase\foreign@title
845   \else
846     \MakeUppercase\local@title
847   \fi}\par
848 \vspace*{3cm}
849 \nohyphens{\@authname\ \@authsur}\par
850 \end{center}
851 \vspace*{2.1cm}
852 \begin{flushright}
853 \begin{minipage}{8.45cm}
854 \frontcover@maintext
855 \end{minipage}\par
856 \vspace*{7.5mm}
857 \nohyphens{%
858 \begin{tabularx}{8.45cm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
859   \local@advisorstring: &
860   \count1=0
861   \toks@={}
862   \@whilenum \count1<\@advisor \do{%
863     \ifcase\count1 % same as \ifnum0=\count1
864       \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
865         \expandafter\endcsname\expandafter\space%
866         \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
867     \else
868       \toks@=\expandafter\expandafter\expandafter{%
869         \expandafter\the\expandafter\toks@%
870         \expandafter&\expandafter\space%
871         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
872         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
873     }%
874   \fi
875   \advance\count1 by 1}
876   \the\toks@
877 \end{tabularx}}\par
878 \end{flushright}
879 \vspace*{\fill}
880 \begin{center}
881 \local@cityname\par
882 \local@monthname\ de \number\year
883 \end{center}
884 \end{titlepage}
885 \global\let\maketitle\relax%
886 \global\let\and\relax}

```

```

887 \newcommand\makefrontpage{%
888   \begin{center}
889     \sloppy\nohyphens{
890       \if@english
891         \MakeUppercase\foreign@title
892       \else
893         \MakeUppercase\local@title
894       \fi}\par
895     \vspace*{7mm}
896     {\@authname\ \@authsur}\par
897 \end{center}}\par
898 \vspace*{4mm}
899 \frontpage@maintext
900 \vspace*{16mm}
901 \nohyphens{%
902 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }}>\raggedright\arraybackslashX@{}
903   \local@advisorstring: &
904   \count1=0
905   \toks@={}
906   \@whilenum \count1<\@advisor \do{%
907     \ifcase\count1 % same as \ifnum0=\count1
908       \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
909         \expandafter\endcsname\expandafter\space%
910         \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
911     \else
912       \toks@=\expandafter\expandafter\expandafter{%
913         \expandafter\the\expandafter\toks@%
914         \expandafter&\expandafter\space%
915         \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
916         \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
917       }%
918     \fi
919     \advance\count1 by 1}
920   \the\toks@
921 \end{tabularx}}\par
922 \vspace*{20mm}
923 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }}>\raggedright\arraybackslashX@{}
924   Aprovada por: &
925   \count1=0
926   \toks@={}
927   \@whilenum \count1<\@examiner \do{%
928     \ifcase\count1 % same as \ifnum0=\count1
929       \toks@=\expandafter{\csname CoppeExaminer:\the\count1%
930         \expandafter\endcsname\expandafter\\}
931     \else
932       \toks@=\expandafter\expandafter\expandafter{%
933         \expandafter\the\expandafter\toks@%
934         \expandafter&\expandafter\space%
935         \csname CoppeExaminer:\the\count1\expandafter\endcsname%
936         \expandafter\space\\
937       }%
938     \fi
939     \advance\count1 by 1}
940   \the\toks@

```



```

941 \end{tabularx}}\par
942 \vspace*{\fill}
943 \frontpage@bottomtext}

944 \newcommand\coppe@hypersetup{%
945 \begin{group}
946 % changes to \toks@ and \count@ are kept local;
947 % it's not necessary for them, but it is usually the case
948 % for \count1, because the first ten counters are written
949 % to the DVI file, thus you got lucky because of PDF output
950 \toks@={}% in this special case not necessary
951 \count@=0 %
952 \@whilenum\count@<\value{keywords}\do{%
953 % * a keyword separator is not necessary,
954 % if there is just one keyword
955 % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
956 % at least once, to get rid of the loop depended \count@
957 \ifcase\count@ % same as \ifnum0=\count@
958 \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
959 \else
960 \toks@=\expandafter\expandafter\expandafter{%
961 \expandafter\the\expandafter\toks@
962 \expandafter;\expandafter\space
963 \csname CoppeKeyword:\the\count@\endcsname
964 }%
965 \fi
966 \advance\count@ by 1 %
967 }%
968 \edef\x{\endgroup
969 \noexpand\hypersetup{%
970 pdfkeywords={\the\toks@}%
971 }%
972 }%
973 \x
974 \hypersetup{%
975 pdfauthor={\@authname\ \@authsur},
976 pdftitle={\local@title},
977 pdfsubject={\local@doctype\ de \@degree\ em \local@deptname\ da COPPE/UFRJ},
978 pdfcreator={LaTeX with CoppeTeX toolkit},
979 breaklinks={true},
980 raiselinks={true},
981 pageanchor={true},
982 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il.” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain $\TeX$ command `\isundefined` [?].

```

983 \newcommand\makecatalog{%
984 \vspace*{\fill}
985 \begin{center}
986 \setlength{\fboxsep}{5mm}
987 \framebox[120mm][c]{\makebox[5mm][c]{}}%
988 \begin{minipage}[c]{105mm}

```

```

989 \setlength{\parindent}{5mm}
990 \noindent\sloppy\nohyphens\@authsurn,
991 \nohyphens\@authname\par
992 \nohyphens{%
993 \if@english
994 \foreign@title%
995 \else
996 \local@title%
997 \fi/\@authname\ \@authsurn. -- \local@cityname:
998 UFRJ/COPPE, \number\year.}\par
999 \pageref{front:pageno},
1000 \pageref{LastPage}
1001 p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm.\par
1002 % There is an issue here. When the last entry must be split between lines,
1003 % the spacing between it and the next paragraph becomes smaller.
1004 % Should we manually introduce a fixed space? But how could we know that
1005 % a name was split? Is this happening yet?
1006 \nohyphens{%
1007 \begin{tabularx}{100mm}[b]{@{}l@{ }}>\raggedright\arraybackslashX@{}}
1008 \local@advisorstring: &
1009 \count1=0
1010 \toks@={}
1011 \@whilenum \count1<\@advisor \do{%
1012 \ifcase\count1 % same as \ifnum0=\count1
1013 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1014 \expandafter\endcsname\expandafter\space%
1015 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1016 \else
1017 \toks@=\expandafter\expandafter\expandafter{%
1018 \expandafter\the\expandafter\toks@
1019 \expandafter&\expandafter\space
1020 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1021 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1022 }%
1023 \fi
1024 \advance\count1 by 1}
1025 \the\toks@
1026 \end{tabularx}}\par
1027 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
1028 UFRJ/COPPE/Programa de \local@deptname, \number\year.}\par
1029 Refer{\^ e}ncias Bibliogr{' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
1030 \count1=0
1031 \count2=1
1032 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
1033 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
1034 \advance\count1 by 1
1035 \advance\count2 by 1}
1036 I. \csname CoppeAdvisorSurname:0\endcsname,%
1037 \ \csname CoppeAdvisorName:0\endcsname%
1038 \ifthenelse{\@advisor>1}{\ \emph{et al.}}{}}.
1039 II. \local@universityname, COPPE, Programa de \local@deptname.
1040 III. T{' i}tulo.}
1041 \end{minipage}}
1042 \end{center}

```

```

1043 \vspace*{\fill}}

\dedication
1044 \newcommand\dedication[1]{
1045 \gdef\@dedic{#1}
1046 \cleardoublepage
1047 \vspace*{\fill}
1048 \begin{flushright}
1049 \begin{minipage}{60mm}
1050 \raggedleft \it \normalsize \@dedic
1051 \end{minipage}
1052 \end{flushright}}

abstract (env.) This is a specialization of the abstract in the article standard class.
1053 \newenvironment{abstract}{%
1054 \clearpage
1055 \thispagestyle{plain}
1056 \abstract@toptext\par
1057 \vspace*{8.6mm}
1058 \begin{center}
1059 \sloppy\nohyphens{\MakeUppercase\local@title}\par
1060 \vspace*{13.2mm}
1061 \@authname\ \@authsur \par
1062 \vspace*{7mm}
1063 \local@monthname/\number\year
1064 \end{center}\par
1065 \vspace*{\fill}
1066 \noindent%
1067 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}
1068 \local@advisorstring: &
1069 \count1=0
1070 \toks@={}
1071 \@whilenum \count1<\@advisor \do{%
1072 \ifcase\count1 % same as \ifnum0=\count1
1073 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1074 \expandafter\endcsname\expandafter\space%
1075 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1076 \else
1077 \toks@=\expandafter\expandafter\expandafter{%
1078 \expandafter\the\expandafter\toks@
1079 \expandafter&\expandafter\space
1080 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1081 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1082 }%
1083 \fi
1084 \advance\count1 by 1}
1085 \the\toks@
1086 \end{tabularx}\par
1087 \vspace*{2mm}
1088 \noindent\local@deptstring: \local@deptname\par
1089 \vspace*{7mm}}{\vspace*{\fill}}

foreignabstract (env.)

```

```

1090 \newenvironment{foreignabstract}{%
1091   \clearpage
1092   \thispagestyle{plain}
1093   \begin{otherlanguage}{english}
1094   \foreignabstract@toptext\par
1095   \vspace*{8.6mm}
1096   \begin{center}
1097     \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
1098     \vspace*{13.2mm}
1099     \@authname\ \@authsur \par
1100     \vspace*{7mm}
1101     \foreign@monthname/\number\year
1102   \end{center}\par
1103   \vspace*{\fill}
1104   \noindent%
1105   \begin{tabularx}{\textwidth}[b]{\@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1106     \foreign@advisorstring: &
1107     \count1=0
1108     \toks@={}
1109     \@whilenum \count1<\@advisor \do{%
1110       \ifcase\count1 % same as \ifnum0=\count1
1111         \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1112           \expandafter\endcsname\expandafter\space%
1113           \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1114         \else
1115           \toks@=\expandafter\expandafter\expandafter{%
1116             \expandafter\the\expandafter\toks@
1117             \expandafter&\expandafter\space
1118             \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1119             \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1120           }%
1121         \fi
1122         \advance\count1 by 1}
1123     \the\toks@
1124   \end{tabularx}\par
1125   \vspace*{2mm}
1126   \noindent\foreign@deptstring: \foreign@deptname\par
1127   \vspace*{7mm}}{%
1128   \end{otherlanguage}
1129   \vspace*{\fill}
1130   \global\let\@author\@empty
1131   \global\let\@date\@empty
1132   \global\let\foreign@title\@empty
1133   \global\let\foreign@title\relax
1134   \global\let\local@title\@empty
1135   \global\let\local@title\relax
1136   \global\let\author\relax
1137   \global\let\author\relax
1138   \global\let\date\relax}

```

\listoffigures

```

1139 \renewcommand\listoffigures{%
1140   \coppe@hasLof
1141   \if@twocolumn

```

```

1142     \@restonecoltrue\onecolumn
1143     \else
1144     \@restonecolfalse
1145     \fi
1146     \chapter*{\listfigurename}%
1147     \addcontentsline{toc}{chapter}{\listfigurename}%
1148     \@mkboth{\MakeUppercase\listfigurename}%
1149     {\MakeUppercase\listfigurename}%
1150     \@starttoc{lof}%
1151     \if@restonecol\twocolumn\fi
1152     }

\listoftables
1153 \renewcommand\listoftables{%
1154     \if@twocolumn
1155     \@restonecoltrue\onecolumn
1156     \else
1157     \@restonecolfalse
1158     \fi
1159     \chapter*{\listtablename}%
1160     \addcontentsline{toc}{chapter}{\listtablename}%
1161     \@mkboth{%
1162         \MakeUppercase\listtablename}%
1163     {\MakeUppercase\listtablename}%
1164     \@starttoc{lot}%
1165     \if@restonecol\twocolumn\fi
1166     }

\printlosymbols
1167 \newcommand\printlosymbols{%
1168 \renewcommand\glossaryname{\listsymbolname}%
1169 \@input@{\jobname.los}}

\makelosymbols
1170 \def\makelosymbols{%
1171     \newwrite\@losfile
1172     \immediate\openout\@losfile=\jobname.syx
1173     \newcommand\symb1[3][\@bsphack\begingroup
1174     \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}}%
1175     \@sanitize%
1176     \@wrlos{\@tempsymb1}{##2}{##3}\typeout%
1177     {Writing index of symbols file \jobname.syx}%
1178     \let\makelosymbols\empty%
1179 }%
1180 \@onlypreamble\makelosymbols

1181 \AtBeginDocument{%
1182 \@ifpackageloaded{hyperref}{%
1183     \newcommand\@wrlos[3]{%
1184         \protected@write\@losfile{%
1185             {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
1186             \endgroup%
1187             \@esphack}}}%
1188     \newcommand\@wrlos[3]{%

```

```

1189 \protected@write\@losfile{}%
1190 {\string\indexentry{#1[#2] #3}{\thepage}}%
1191 \endgroup%
1192 \@esphack}}}%

```

\printloabbreviations

```

1193 \newcommand\printloabbreviations{%
1194 \renewcommand\glossaryname{\listabbreviationname}%
1195 \@input@{\jobname.lab}}

```

\makeloabbreviations

```

1196 \def\makeloabbreviations{%
1197 \newwrite\@labfile
1198 \immediate\openout\@labfile=\jobname.abx
1199 \newcommand\abbrev[3][\@bsphack\beginingroup
1200 \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
1201 \@sanitize
1202 \@wrlab{\@tempsymb1}{##2}{##3}}\typeout
1203 {Writing index of abbreviations file \jobname.abx}%
1204 \let\makeloabbreviations\@empty
1205 }
1206 \@onlypreamble\makeloabbreviations

1207 \AtBeginDocument{%
1208 \@ifpackageloaded{hyperref}{%
1209 \newcommand\@wrlab[3]{%
1210 \protected@write\@labfile{}%
1211 {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
1212 \endgroup%
1213 \@esphack}}{%
1214 \newcommand\@wrlab[3]{%
1215 \protected@write\@labfile{}%
1216 {\string\indexentry{#1[#2] #3}{\thepage}}%
1217 \endgroup%
1218 \@esphack}}}%

1219 %%% \AtBeginDocument{%
1220 %%% \@ifpackageloaded{hyperref}{%
1221 %%% \def\@wrlab#1#2{%
1222 %%% \protected@write\@labfile{}%
1223 %%% {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
1224 %%% \endgroup
1225 %%% \@esphack}}{%
1226 %%% \def\@wrlab#1#2{%
1227 %%% \protected@write\@labfile{}%
1228 %%% {\string\indexentry{[#1] #2}{\arabic{page}}}%
1229 %%% \endgroup
1230 %%% \@esphack}}

1231 % Some macros used to generate cataloging information.
1232 \AtBeginDocument{%
1233 \ltx@ifpackageloaded{hyperref}{%
1234 \def\coppe@bibEnd{%
1235 \immediate\write\@auxout{%
1236 \string\newlabel{bib:end}{\arabic{page}}{\page.\arabic{page}}}}}%

```

```

1237 \def\coppe@bibBegin{%
1238 \immediate\write\@auxout{%
1239 \string\newlabel{bib:begin}{\arabic{page}}{page.\arabic{page}}}%
1240 \def\coppe@mainBegin{%
1241 \immediate\write\@auxout{%
1242 \string\newlabel{front:pageno}{\Roman{page}}{page.\roman{page}}}%
1243 \def\coppe@hasLof{%
1244 \immediate\write\@auxout{%
1245 \string\newlabel{cat:lofflag}{:~il.;}{page.\roman{page}}}%
1246 }%
1247 \def\coppe@bibEnd{%
1248 \immediate\write\@auxout{%
1249 \string\newlabel{bib:end}{\arabic{page}}}%
1250 \def\coppe@bibBegin{%
1251 \immediate\write\@auxout{%
1252 \string\newlabel{bib:begin}{\arabic{page}}}%
1253 \def\coppe@mainBegin{%
1254 \immediate\write\@auxout{%
1255 \string\newlabel{front:pageno}{\Roman{page}}}%
1256 \def\coppe@hasLof{%
1257 \immediate\write\@auxout{%
1258 \string\newlabel{cat:lofflag}{:~il.;}}%
1259 }%
1260 }
1261 % The reference list is now produced by biblatex's \printbibliography (see the
1262 % bibliography setup after babel, above). The former thebibliography
1263 % redefinition and its \bibindent are no longer used and have been removed; the
1264 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
1265 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (env.)

```

1266 \newlength{\recuolongquote}%
1267 \setlength{\recuolongquote}{4cm}%
1268 \newenvironment*{longquote}[1][default]{%
1269 \list{}%
1270 \footnotesize%
1271 \addtolength{\leftskip}{\recuolongquote}%
1272 \item[]%
1273 \singlespacing%
1274 \ifthenelse{\not\equal{#1}{default}}{\itshape\selectlanguage{#1}}{}%
1275 }\endlist}%

1276 \newenvironment{theglossary}{%
1277 \if@twocolumn%
1278 \@restonecoltrue\onecolumn%
1279 \else%
1280 \@restonecolfalse%
1281 \fi%
1282 \@mkboth{\MakeUppercase\glossaryname}%
1283 {\MakeUppercase\glossaryname}%
1284 \chapter*{\glossaryname}%
1285 \addcontentsline{toc}{chapter}{\glossaryname}
1286 \list{}
1287 {\setlength{\listparindent}{0in}%

```

```

1288 \setlength{\labelwidth}{1.0in}%
1289 \setlength{\leftmargin}{1.5in}%
1290 \setlength{\labelsep}{0.5in}%
1291 \setlength{\itemindent}{0in}%
1292 \sloppy}%
1293 {\if@restonecol\twocolumn\fi%
1294 \endlist}
1295 %
1296 \renewenvironment{theindex}{%
1297 \if@twocolumn
1298 \@restonecolfalse
1299 \else
1300 \@restonecoltrue
1301 \fi
1302 \twocolumn[\@makeschapterhead{\indexname}]%
1303 \mkboth{\MakeUppercase\indexname}%
1304 {\MakeUppercase\indexname}%
1305 \thispagestyle{plain}\parindent\z@
1306 \addcontentsline{toc}{chapter}{\indexname}
1307 \parskip\z@ \@plus .3\p@\relax
1308 \columnseprule \z@
1309 \columnsep 35\p@
1310 \let\item\@idxitem}
1311 {\if@restonecol\onecolumn\else\clearpage\fi}
1312 \if@english
1313 \newcommand\listabbreviationname{List of Abbreviations}
1314 \newcommand\listsymbolname{List of Symbols}
1315 \newcommand\glossaryname{Glossary}
1316 \else
1317 \newcommand\listabbreviationname{Lista de Abreviaturas}
1318 \newcommand\listsymbolname{Lista de S{\`i}mbolos}
1319 \newcommand\glossaryname{Gloss{\`a}rio}
1320 \fi
1321 %
1322 \newcommand\local@advisorstring{Orientador}
1323 \newcommand\foreign@advisorstring{Advisor}
1324 \ifthenelse{\boolean{maledoc}}{%
1325 \newcommand\local@approvedname{Examinado por}%
1326 }{%
1327 \newcommand\local@approvedname{Examinada por}%
1328 }
1329 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
1330 \newcommand\local@deptstring{Programa}
1331 \newcommand\foreign@deptstring{Department}
1332 \newcommand\local@cityname{Rio de Janeiro}
1333 \newcommand\local@statename{RJ}
1334 \newcommand\local@countryname{Brasil}
1335 %
1336 \newcommand\frontcover@maintext{
1337 \sloppy\nohyphens{\local@doctype\ de \@degree\name\
1338 \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
1339 ao Programa de P{\`o}s-gradua{\c c}{\~ a}o em \local@deptname,
1340 COPPE, da \local@universityname, como parte dos requisitos
1341 necess{\`a}rios {\`a} obten{\c c}{\~ a}o do t{\`i}tulo de

```



```

1342 \local@degname\ em \local@deptname.}
1343 }
1344 %
1345 \newcommand\frontpage@maintext{
1346 \noindent {\MakeUppercase\local@doctype}
1347 \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
1348 \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
1349 DE P{\~ O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
1350 UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
1351 NECESS{\~ A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
1352 {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
1353 {\MakeUppercase\local@deptname.\par}}%
1354 }
1355 %
1356 \newcommand\frontpage@bottomtext{%
1357 \begin{center}
1358 {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
1359 {\MakeUppercase\local@monthname\ DE \number\year}
1360 \end{center}}%
1361 }
1362 %
1363 \newcommand\abstract@toptext{%
1364 \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
1365 \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
1366 \sloppy\nohyphens{{\~ a} COPPE/UFRJ como parte dos requisitos
1367 necess{\~ a}rios para a obten{\c c}{\~ a}o do grau de
1368 \local@degname\ em Ci{\^ e}ncias (\@degree)}
1369 }
1370 \newcommand\foreignabstract@toptext{%
1371 \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
1372 COPPE/UFRJ as a partial fulfillment of the requirements for the
1373 degree of \foreign@degname\ of Science (\@degree)}
1374 }
1375 %
1376 </class>
1377 <*glossary>
1378 actual '='
1379 quote '!'
1380 level '>'
1381 %%%% delim_0 "", p. "
1382 delim_0 "\\dotfill "
1383 lethead_flag 0
1384 headings_flag 0
1385 preamble
1386 "\n\\begin{theglossary}\n \\makeatletter"
1387 postamble
1388 "\n \\end{theglossary}\n"
1389 </glossary>
1390 \newcommand{\annex}{
1391 \iflanguage{brazilian}
1392 {\renewcommand{\appendixname}{Anexo}}
1393 {\renewcommand{\appendixname}{Annex}}
1394 \appendix

```

Acknowledgments

Thanks to all COPPETEX users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their helpful comments. The authors would like to thank the National Council for Scientific and Technological Development (CNPq) of Brazil.

Index

Numbers written in *italic* refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in roman refer to the code lines where the entry is used.

Symbols	\@wrlos 1176, 1183, 1188	\AtBeginBibliography
\@advisor 748,	_ 77, 307, 440,	... 693, 696, 1264
750, 751, 752,	467, 598, 762,	\AtEndBibliography 1265
753, 754, 755,	804, 849, 882,	\author
862, 906, 1011,	896, 975, 977,	20, <u>765</u> , 1136, 1137
1038, 1071, 1109	997, 1027, 1037,	\autoref .. 171, 194,
\@authname ... 766,	1038, 1061,	209, 242, 260, 297
849, 896, 975,	1099, 1337,	
991, 997, 1061, 1099	1342, 1358,	
\@author 1130	1359, 1365,	B
\@authsur ... 767,	1368, 1371, 1373	\backmatter ... 557, <u>823</u>
849, 896, 975,	\~ 431, 433,	\beta 534, 536
990, 997, 1061, 1099	435, 439, 466,	\bottomrule ... 223,
\@copennumericfalse 659	530, 532, 630,	256, 273, 293, 488
\@copennumerictrue . 660	639, 648, 727,	
\@date 1131	733, 1339, 1341,	C
\@dedic 1045, 1050	1349, 1351, 1367	\c@keywords 781
\@degree 626,		\caption .. 164, 175,
634, 643, 652,	A	199, 213, 246,
785, 786, 1368, 1373	\abbrev 544,	264, 285, 302,
\@degree 627, 635,	545, 546, 547,	439, 466, 515, 525
644, 653, 787,	548, 549, 550, 1199	\captionenglish .. 671
788, 977, 1027, 1337	abstract (env.) ... <u>1053</u>	\cite 151, 448,
\@englishfalse 623	\abstract@top ..	450, 451, 452,
\@englishtrue 624	 1056, 1363	453, 454, 456,
\@examiner	\addbibresource ... 12	457, 458, 460,
. 760, 762, 763, 927	\addto 671	475, 477, 478,
\@labfile 1197,	\addtolength 1271	479, 480, 481,
1198, 1210,	\advisor . 21, 22, 23, <u>748</u>	483, 484, 485, 487
1215, 1222, 1227	\alpha 534, 540	\citep 154, 677
\@losfile 1171,	\annex ... 574, 584, 1390	\cit 154, 448, 450,
1172, 1184, 1189	\appendixname 1392, 1393	451, 452, 453,
\@tempsymb ... 1174,	\arraybackslash ...	455, 456, 457,
1176, 1200, 1202	. 858, 902, 923,	459, 460, 475,
\@wrlab . 1202, 1209,	1007, 1067, 1105	477, 478, 479,
1214, 1221, 1226		480, 482, 483,
		484, 486, 487, 677

<code>\coppe@bibBegin</code> 696, 1237, 1250, 1265	<code>foreignabstract</code> (env.) 1090	<code>\local@deptname</code> 703, 706, 709,
<code>\coppe@bibEnd</code> . 700, 1234, 1247, 1265	<code>\foreignabstract@toptext</code> 1094, 1370	712, 715, 718, 721, 724, 727,
<code>\coppe@hasLof</code> 1140, 1243, 1256	<code>\foreigntitle</code> .. 19, 745	730, 733, 736, 739, 977, 1028,
<code>\coppe@hypersetup</code> 832, 944	<code>\freeconfig</code> 784	1039, 1088, 1339, 1342, 1353
<code>\coppe@mainBegin</code> 818, 1240, 1253	<code>\frontcover@maintext</code> 854, 1336	<code>\local@deptstring</code> 1088, 1330
<code>\coppe@orig@printbibliography</code> 943, 1356	<code>\frontmatter</code> 39, 801, 806	<code>\local@doctype</code> 630, 639,
<code>\CoppeTeX</code> 584, 614	<code>\frontpage@bottomtext</code> 899, 1345	648, 656, 793, 794, 977, 1027, 1337, 1346, 1365
D		
<code>\date</code> 31, 769, 1138	<code>\glossaryname</code> . 1168, 1194, 1282,	<code>\local@monthname</code> .. 773, 882, 1063, 1359
<code>\dedication</code> ... 40, 1044	1283, 1284,	<code>\local@statename</code> 1333, 1358
<code>\defbibheading</code> 688	1285, 1315, 1319	<code>\local@title</code> 743, 846, 893, 976, 996, 1059, 1134, 1135
<code>\department</code> 30, 701		<code>\local@universityname</code> .. 1039, 1329, 1340
E		
<code>\endfirsthead</code> 304	<code>\if@coppenumeric</code> 659, 678	<code>longquote</code> (env.) .. 1266
<code>\endfoot</code> 312	<code>\if@english</code> 623, 666, 843, 890, 993, 1312	
<code>\endhead</code> 309	<code>\if@openright</code> 824	
<code>\endlastfoot</code> 317	<code>\iflanguage</code> 1391	
environments:	<code>\ifstrempty</code> . 1174, 1200	
<code>abstract</code> 1053	<code>\itshape</code> 1274	
<code>foreignabstract</code> 1090		
<code>longquote</code> 1266		
<code>\examiner</code> 25, 26, 27, 28, 29, 760		
F		
<code>\footnote</code> 112, 118	<code>\keyword</code> . 33, 34, 35, 779	
<code>\foreign@advisorstring</code> .. 757, 1106, 1323		
<code>\foreign@degname</code> .. . 629, 637, 646, 655, 791, 792, 1373	<code>\leftskip</code> 1271	
<code>\foreign@deptname</code> 704, 707, 710, 713, 716, 719, 722, 725, 728, 731, 734, 737, 740, 1126	<code>\listabbreviationname</code> .. 1194, 1313, 1317	
<code>\foreign@deptstring</code> 1126, 1331	<code>\listoffigures</code> 59, 1139	
<code>\foreign@doctype</code> .. . 631, 640, 649, 657, 795, 796, 1371	<code>\listoftables</code> . 60, 1153	
<code>\foreign@monthname</code> 776, 1101	<code>\listsymbolname</code> 1168, 1314, 1318	
<code>\foreign@title</code> 746, 844, 891, 994, 1097, 1132, 1133	<code>\local@advisorstring</code> . 756, 859, 903, 1008, 1068, 1322	
	<code>\local@approvedname</code> 1325, 1327	
	<code>\local@cityname</code> ... 881, 997, 1332, 1358	
	<code>\local@countryname</code> 1334, 1358	
	<code>\local@degname</code> 628, 636, 645, 654, 789, 790, 1342, 1352, 1368	
		<code>\newif</code> 623, 659
		<code>\newlength</code> 1266
		<code>\node</code> 505, 508
		<code>\not</code> 1274
		N
		<code>\printbibliography</code> 559, 692, 695, 697, 698, 1261
		<code>\printloabbreviations</code> 62, 1193
		<code>\printlosymbols</code> 61, 1167
		<code>\providecommand</code> 785, 787, 789, 791, 793, 795
		P

Q	S	T		
<code>\quote</code> 140	<code>\section</code> 95, 106, 114, 138, 169, 207, 227, 532	<code>\tablename</code> 307 <code>\tableofcontents</code> . . 58 <code>\thetable</code> 307 <code>\title</code> 18, <u>742</u> <code>\toprule</code> 216, 250, 268, 288, 474		
R	<code>\selectlanguage</code> . . 1274 <code>\singlespacing</code> . . . 1273 <code>\subsection</code> 238 <code>\symbl</code> 131, 132, 535, 536, 537, 538, 539, 540, 541, 1173			
<code>\raggedright</code> 858, 902, 923, 1007, 1067, 1105 <code>\recuolongquote</code> 1266, 1267, 1271 <code>\resizebox</code> . . . 260, 266		<th>U</th> <td><code>\url</code> 71</td>	U	<code>\url</code> 71

Change History

v0.0	General: Creation Date. 1	v2.2	General: Matching new guidelines, including new logo. 1
v0.1	General: Documentation: bibliography fixed, title translation. 1 Sourceforge submission. 1	v2.2.2	General: Fixed some text constants in .bib and documented it here. 1
v0.2	General: Unification of the code for the list of symbols and abbreviations. 1	v3.0	General: Added support for monographs in English. 21 New approval page layout. . . . 26 <code>\department</code> : Added new course on Nanotechnology. 21 <code>\examiner</code> : Examiners expansion without degree. 22
v0.3	General: Added 'draft' option. . . . 1 <code>\maketitle</code> : Added number of examiners test. 24 Generalization. 24	v3.1	<code>\department</code> : Included a sort key in <code>\symbl</code> 21
v0.4	General: Beta documentation. . . . 1	v3.2	General: Fixed version problem between <code>cls</code> and <code>dtx</code> 1
v0.5	<code>abstract</code> : Changed from macro to environment. 28 <code>\backmatter</code> : Added mainmatter pages counter. 23 <code>foreignabstract</code> : Changed from macro to environment. 29	v3.3	General: Extend <code>abrev</code> to work like <code>\symb</code> and accept a sorting key . 1
v1.0	General: First COPPE _{TEX} release. . . 1	v3.4	General: Some examples for figures, tables, longtables, etc. . 1
v2.0	General: COPPE _{TEX} release 2.0. . . 1	v3.4.1	General: Fix english option to consider brazilian a second language, describe it in document 21
v2.1	General: COPPE _{TEX} release 2.1: Matching the new rules. 1 <code>\advisor</code> : Advisors, co-advisors, co-co-advisors, etc., all of them are simply considered advisors. 22	v3.5	General: New command <code>annex</code> . . 35
		v3.5.1	General: New command <code>annex</code> now supports brazilian or english . 35